

SOFTWARE ENGINEERING

Lab3:

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Advitha-Shetty/37_climber

Project: Climber- Pygame vertical platform

Tasks1:

I am working on Task 1 of my Climber Repair Lab. Analyze the existing game.py code I provided.

The README says the Height shown in the HUD must track the best/highest height the player has reached during the current run and must never decrease when the player falls or moves downward.

Identify the exact bug in the current height calculation and explain the smallest safe code change needed to fix it.

Do not implement Tasks 2, 3, or 4 yet. Do not change any unrelated game behavior. After explaining the fix, show only the relevant corrected code section.

Tasks2:

I am working on Task 2 of my Climber Repair Lab.

Analyze the existing game.py code I provided and the README requirements.

Task 2 requirement:

Implement `platform_color(index, total)` so that platforms gradually change color as the player climbs higher. `index=0` is the ground platform, and `total` is the total number of generated platforms.

Requirements:

- Return an RGB tuple `(r, g, b)` for platforms whose color should be changed.
- The color should gradually change based on the platform's index relative to the total number of platforms.
- The ground platform should remain the normal/default green.
- Keep the existing `Platform` class and other game behavior unchanged.

- Do not implement Task 3 or Task 4 yet.
- Make the smallest reasonable change needed for Task 2.
- Explain the code before showing the exact function I should replace.

Tasks3:

I am working on Task 3 of my Climber Repair Lab.

Analyze the existing `game.py` code and README requirements.

Task 3 requirement:

Implement `moving_platform_speed(index, total)` so that some platforms move horizontally while others remain static.

The existing `Platform.update()` method already handles horizontal oscillation using the speed returned by this function, so do not rewrite that movement logic.

Requirements:

- Return a small horizontal speed in pixels per frame for platforms that should move.
- Return 0 or None for platforms that should remain static.
- Use the platform index to determine which platforms move.
- Keep the ground platform static.
- Make the movement clearly visible when playing the game.
- Keep all existing Task 1 and Task 2 behavior unchanged.
- Do not implement Task 4 yet.
- Make the smallest reasonable code change needed for Task 3.
- Explain the change and show the exact function I should replace.

Tasks4:

I am working on Task 4 of my Climber Repair Lab.

Analyze the existing `game.py` code and README requirements.

Task 4 requirement:

Implement `on_coin_collected(coin, score)`, which is called immediately after a coin is collected and its value has already been added to the score.

Requirements:

- Add a visible coin-collection effect such as a small sparkle or visual feedback.
- Do not change the existing coin value or scoring logic.
- Do not break the existing coin removal behavior.
- Keep Tasks 1, 2, and 3 unchanged.
- The effect should be visible when the player collects a coin.

- Do not add unnecessary libraries or external assets.
- Make the smallest reasonable change needed.
- Do not modify unrelated game behavior.
- Explain exactly what needs to change and show the complete code sections that need to be replaced or added.